

Finite algebraic envelopes and the Boone–Higman conjecture

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Abstract

We prove the Boone–Higman conjecture. A finitely generated group has decidable word problem if and only if it embeds in a finitely presented simple group.

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1 Introduction

The word problem for a group with a specified finite generating set asks whether a given word in the generators and their inverses represents the identity. It is *decidable* if an algorithm answers this question for every word. A group is *simple* here if it is nontrivial and its only normal subgroups are the identity subgroup and the whole group.

Theorem 1.1. *For every finitely generated group G , the following are equivalent:*

- (i) G has decidable word problem.
- (ii) G embeds in a finitely presented simple group.

An embedding throughout means an injective group homomorphism. In the constructive implication, G is supplied with a finite generating set and a word-problem decider; it need not be finitely presented. Theorem 1.1 resolves the Boone–Higman conjecture positively.

Higman’s embedding theorem characterizes the finitely generated subgroups of finitely presented groups as those admitting a recursively enumerable set of defining relations [10]; see also [1, Theorem A]. Decidability of the word problem is stronger: it supplies an algorithm for both equality and inequality. Boone and Higman characterized this condition by the existence of embeddings

$$G \leq H \leq K, \quad H \text{ simple, } K \text{ finitely presented}$$

[6, Theorem I]. They asked whether the simple overgroup H could itself be chosen finitely presented [6, p. 43]. The two properties in their theorem belong to different groups, and finite presentation does not in general pass to subgroups. Thompson subsequently obtained a finitely generated simple overgroup with decidable word problem [18]; a later proof appears in [9, Theorem 7.1]. This strengthens the conclusion to finite generation of the simple overgroup, but does not supply its finite presentation.

Positive cases of the conjecture include hyperbolic and contracting self-similar groups [4, Theorem A and Corollary D], Baumslag–Solitar groups and finite-rank free-by-cyclic groups [8, Theorem A], and automorphism groups of finite-rank free groups and mapping class groups of orientable punctured surfaces of finite type [5, Theorem A and Corollary B]. The survey [3] describes further results and the development of the conjecture.

Our principal intermediate result concerns permutation actions. An action on a set X is *faithful* if only the identity acts trivially, and *two-transitive* if it is transitive on the ordered pairs of distinct points of X .

Theorem 1.2. *Every finitely generated group with decidable word problem embeds in a finitely presented group Γ admitting a faithful two-transitive action on a countably infinite set. Every point stabilizer of this action is finitely generated.*

The action is designed for the twisted Brin–Thompson construction of Belk and Zaremsky, which embeds an acting group in a simple group [2, Section 3 and Theorem 3.4]. Their original finite-presentation theorem used finitely many orbits on finite subsets of each size and finitely presented finite-set stabilizers [2, Theorem D]. Zaremsky subsequently showed that finite presentation of the acting group, finite generation of point stabilizers, and finitely many orbits on unordered pairs suffice [20, Theorem A and Corollary B]. Two-transitivity supplies the last condition here. We state this criterion precisely in Theorem 2.3; the construction below provides its faithful action and the embedding of the prescribed group.

The proof

The action is affine. Translations move the first point of an ordered pair to zero, so two-transitivity reduces to transitivity of the linear part on nonzero vectors. A simple algebra A provides the required coefficient operations: if $0 \neq m \in A$, then its identity belongs to AmA , and hence is a finite sum of left and right multiples of m . Elementary row operations can use this identity to move vectors. Finite presentation, however, will come from a different algebra.

Over $k = \mathbb{F}_2$, we construct a finitely presented algebra B , a simple subalgebra $A \subseteq B$ with its own identity, and an injective nonunital map $\phi : B \rightarrow B$ with image in A . The additive direct limit $M_\infty = \varinjlim (B, \phi)$ is the carrier for the action on $X = M_\infty^6$. Every finite collection of its coordinates lies in one copy of B ; advancing once places them in A . Thus simplicity becomes available at a later stage without requiring the finitely presented algebra B itself to be simple.

The coefficient ring $R = B \otimes_k B^{\text{op}}$ acts on B by left and right multiplication. We give an explicit finite presentation of the affine Steinberg group

$$\Gamma_0 = B^6 \rtimes \text{St}_6(R).$$

The Steinberg group maps onto the elementary matrix group $E_6(R)$, but its kernel is an obstruction to the desired faithful affine action. We remove that entire kernel by a single application of the coefficient map induced by ϕ . The key algebraic input consists of two splitting pairs commuting with the coefficient image. Their corner identities give an exact calculation in the Steinberg presentation; no centrality hypothesis on the kernel is needed.

The resulting endomorphism of Γ_0 defines a finitely presented mapping torus Γ . In its direct-limit description the Steinberg kernel has disappeared, leaving elementary matrices acting on M_∞^6 , together with translations and a stage shift. Simplicity of A gives the nonzero-vector orbit required for two-transitivity. Faithfulness uses two further properties of the algebra construction: a separating vector detects every nonzero coefficient after one advance, and the proper support $\phi(1_B) \neq 1_B$ makes the stage shift detectable. The stabilizer of zero is the quotient by the translation subgroup, so it is finitely generated. To embed the original G , we first use an HNN extension to express its elements as controlled commutators; balanced elementary diagonals then place this copy of G in the linear part.

It remains to construct the algebras with these simultaneous properties. We do so through a finitely presented monoid and a decidable monoid that encode one another. A finite presentation is compiled syntactically from a program index, before any totality assertion is available. Its partial evaluator controls the length of every call to the encoded program. Long codes make the calls needed to normalize a word strictly shorter than that word. A fixed-point argument and induction on length then establish totality and compatibility with multiplication. Finite-tuple selectors isolate individual basis terms in a contracted monoid algebra, producing A ; a separate separator distinguishes ordered pairs of basis terms, producing the faithful coefficient detector.

The finite-presentation construction follows the machine-encoding tradition of Post and Murskii [15, 17]. Here the bounded evaluator permits the two monoids to be constructed together. The selectors strengthen the contextual-collapse property in the semigroup counterpart of the Boone–Higman theorem [6, pp. 49–52]: isolating each term of a finite linear combination gives the simplicity needed for the affine action.

Section 2 states the algebraic data and gives the HNN reduction. Sections 3–5 construct the action and prove the two theorems from those data. Section 6 constructs the algebras from a precise monoid theorem, proved by bounded self-reference in Section 7. Appendix A gives the full finite compiler and its evaluator proof.

2 The group input and the algebraic data

We first enlarge the input group so that each of its elements is a commutator in the larger group. This will allow us to embed the input group in an elementary matrix group in Section 5. We then state the algebraic construction used in the next three sections. All rings and algebras have identities; a homomorphism explicitly called nonunital is not required to preserve them. Our commutator convention is $[a, b] = aba^{-1}b^{-1}$.

Lemma 2.1. *Let G be finitely generated with decidable word problem. There is a finitely generated group $\widehat{G}$ with decidable word problem containing $G \times G$, and an element $s \in \widehat{G}$, such that*

$$[(g, g), s] = (g, 1) \quad (g \in G).$$

Proof. Take the HNN extension

$$\widehat{G} = \langle G \times G, s \mid s(g, g)s^{-1} = (1, g) \quad (g \in G) \rangle.$$

The associated subgroups are the diagonal and the second factor, with the indicated isomorphism. Britton’s Lemma embeds the base [7]; see also the precise effective HNN formulation in [6, p. 44].

We spell out the decision procedure. Store each base word as a pair of words on the given generators of G . Membership of (a, b) in the diagonal is decided by testing $ab^{-1} = 1$, and membership in the second factor by testing $a = 1$. Replace each available pinch by

$$s(a, a)s^{-1} \mapsto (1, a), \quad s^{-1}(1, b)s \mapsto (b, b),$$

and multiply adjacent base words. Each pinch removes two stable letters. When no pinch remains, a word still containing a stable letter is nontrivial by Britton’s Lemma. Otherwise decide the resulting base word coordinatewise. Thus the procedure terminates and decides the word problem, without requiring a finite presentation of G . The two copies of a finite generating set, together with s , generate $\widehat{G}$. Finally,

$$[(g, g), s] = (g, g)s(g^{-1}, g^{-1})s^{-1} = (g, g)(1, g^{-1}) = (g, 1).$$

This includes finite groups and the trivial group; in the latter case $\widehat{G}$ is infinite cyclic. □

Here is the algebraic construction. A simple algebra means a nonzero algebra with no proper nonzero two-sided ideal. A subalgebra with identity e need not contain the identity of the ambient algebra; its units are taken relative to e . For an idempotent $q \in B$, the subalgebra $qBq = \{qbq : b \in B\}$ is called a *corner*; its identity is q .

Theorem 2.2. *For every finitely generated group L with decidable word problem, there are the following data over $k = \mathbb{F}_2$:*

- (i) *A nonzero finitely presented unital algebra B , a simple subalgebra $A \subseteq B$ with identity e , and an embedding $L \hookrightarrow A^\times$.*
- (ii) *An injective nonunital algebra homomorphism $\phi : B \rightarrow B$ with $\phi(B) \subseteq A$ and $p = \phi(1_B) \neq 1_B$.*
- (iii) *Elements $s_0, s_1, t_0, t_1 \in pBp$ commuting with $\phi(B)$ and satisfying*

$$t_i s_j = \delta_{ij} p, \quad s_0 t_0 + s_1 t_1 = p \quad (i, j \in \{0, 1\}).$$

- (iv) *An element $b_* \in B$ with the following property. Put*

$$R = B \otimes_k B^{\text{op}}, \quad \psi = \phi \otimes \phi^{\text{op}}, \quad (a \otimes b^{\text{op}}) \cdot m = amb.$$

Then

$$r \neq 0 \implies \psi(r) \cdot b_* \neq 0 \quad (r \in R).$$

Theorem 2.2 is proved in Section 6, using the monoid construction of Section 7 and Appendix A. To see how these data produce an action, regard the elements of R as coefficients for row operations on B^6 . Such a coefficient acts on a coordinate by a finite sum of left and right multiplications. Simplicity of A will let a nonzero coordinate produce any prescribed element of A ; applying ϕ places all coordinates in that subalgebra. Finite presentation of B supplies a finitely presented Steinberg group acting by row operations. The splitting pairs make its matrix kernel disappear under the induced coefficient map, while b_* detects every remaining nonidentity matrix after that map is applied. Finally, the support identity $p \neq 1_B$ makes successive vector stages strictly larger, so that shifting stages gives a nontrivial permutation. Sections 3–5 establish these claims and assemble the action.

In particular, $p\phi(b) = \phi(b)p = \phi(b)$, so $1_B \notin \phi(B)$. The detection condition concerns the action after applying ψ ; it does not require the original action of R on B to be faithful.

Apply the theorem with $L = \widehat{G}$ from Lemma 2.1. For the remainder of Sections 3–5, fix these data. Tensor products are over k . Besides p , define

$$P = p \otimes p^{\text{op}}, \quad U_i = s_i \otimes p^{\text{op}}, \quad V_i = t_i \otimes p^{\text{op}}.$$

Tensoring injective linear maps over a field shows that ψ is injective, so $P \neq 0$. Also $P \neq 1_R$, since the nonzero tensor $(1_B - p) \otimes 1_B$ annihilates P on the left. The elements U_i, V_i lie in PRP , commute with $\psi(R)$, and satisfy

$$V_i U_j = \delta_{ij} P, \quad U_0 V_0 + U_1 V_1 = P. \tag{1}$$

These identities follow directly from the corresponding identities in B ; the second tensor factor is always the idempotent p^{op} .

The final passage from an action to a simple group uses the following established result, in the precise form needed here.

Theorem 2.3 (Zaremsky). *Suppose a finitely presented group Λ acts faithfully on a nonempty countable set X . If the point stabilizers are finitely generated and there are finitely many orbits on the unordered two-element subsets of X , then Λ embeds in a nontrivial finitely presented simple group.*

These are the hypotheses of a type-(A) action in [20, Theorem A and Corollary B]. Finite presentation follows from that theorem; the canonical embedding and simplicity are those of the twisted Brin–Thompson construction [2, Section 3 and Theorem 3.4]. Since X is nonempty, the constructed group contains Thompson’s group V on a coordinate and is nontrivial. Our construction will provide a countably infinite X with one orbit on pairs, and will prove each of the other hypotheses separately.

3 A finitely presented affine group

We now use the algebraic data in Theorem 2.2 to construct a finitely presented group of translations and elementary operations. The translations form the additive group of B^6 . We first give their finite presentation relative to a Steinberg group, and then construct the endomorphism used in the next two sections.

3.1 The coefficient ring and the translation module

Put

$$R = B \otimes_k B^{\text{op}}, \quad (a \otimes b^{\text{op}}) \cdot m = amb \quad (m \in B).$$

Thus $r = \sum_{\nu} a_{\nu} \otimes b_{\nu}^{\text{op}}$ acts on B by the operation $m \mapsto \sum_{\nu} a_{\nu} m b_{\nu}$. The opposite multiplication in the second factor makes composition of these operations agree with multiplication in R , so B is a left R -module. The ring R is finitely presented as a unital associative ring: take a finite algebra presentation of B , a second copy with products reversed, the relations commuting every generator of the first copy with every generator of the second, and $2 \cdot 1 = 0$. The universal property of the tensor product identifies the resulting ring with R . The module B is generated by 1_B ; we next show that the relations on this module generator also have a finite description.

Lemma 3.1. *Let Y be a finite algebra generating set for B . For $b \in B$ put*

$$d_b = b \otimes 1 - 1 \otimes b^{\text{op}}, \quad J = \sum_{y \in Y} R d_y.$$

Evaluation at 1_B induces an isomorphism of left R -modules $R/J \cong B$.

Proof. Evaluation is a surjective R -module homomorphism, and every d_y annihilates 1_B . The identity

$$d_{bc} = (b \otimes 1) d_c + (1 \otimes c^{\text{op}}) d_b \tag{2}$$

shows, together with additivity and $d_1 = 0$, that $d_b \in J$ for all $b \in B$. Indeed, B is generated by Y over the prime field k . Consequently

$$a \otimes b^{\text{op}} \equiv ab \otimes 1 \pmod{J} \quad (a, b \in B).$$

Every element of R is therefore congruent modulo J to its evaluation tensored with 1. The additive map $b \mapsto (b \otimes 1) + J$ is an inverse to the induced evaluation map. Thus the kernel is exactly the stated left ideal. $\square$

For a unital ring R and $n \geq 3$, the Steinberg group $\text{St}_n(R)$ has generators $x_{ab}(r)$, where $1 \leq a \neq b \leq n$ and $r \in R$, with relations

$$x_{ab}(r) x_{ab}(s) = x_{ab}(r + s), \tag{3}$$

$$[x_{ab}(r), x_{bc}(s)] = x_{ac}(rs) \quad (a, b, c \text{ distinct}), \tag{4}$$

$$[x_{ab}(r), x_{cd}(s)] = 1 \quad (a \neq d, b \neq c). \tag{5}$$

Let e_{ab} denote the matrix unit. The map $x_{ab}(r) \mapsto I + re_{ab}$ gives a surjection

$$\pi_n : \text{St}_n(R) \longrightarrow E_n(R),$$

where $E_n(R)$ is the subgroup of $\text{GL}_n(R)$ generated by these elementary matrices. Krstić and McCool prove that $\text{St}_n(R)$ is finitely presented whenever R is a finitely presented unital associative ring and $n \geq 4$ [13, Theorem 3]. Their theorem permits noncommutative rings, and applies to the ring constructed above.

3.2 A finite relative presentation

The next lemma supplies the additional relations needed for the translations. Finite presentation of the module and of the linear group alone would not establish its conclusion; we prove the semidirect-product presentation explicitly.

Lemma 3.2. *Let R be a finitely generated unital associative ring, let $J = \sum_{d \in D} Rd$ for a finite subset $D \subset R$, and let $n \geq 4$. If $\text{St}_n(R)$ is finitely presented, then*

$$(R/J)^n \rtimes \text{St}_n(R)$$

is finitely presented, where the action on columns is induced by π_n . More precisely, choose a finite ring generating set $F \subset R$ containing 1. A finite presentation is obtained from a finite presentation of $\text{St}_n(R)$ by adjoining $p_1, \dots, p_n$ and the relations

$$[p_a, p_b] = 1, \tag{6}$$

$$[x_{ab}(s), p_c] = 1 \quad (s \in F, c \neq b), \tag{7}$$

$$[x_{ab}(1), p_b] = p_a, \tag{8}$$

$$[x_{ab}(d), p_b] = 1 \quad (d \in D). \tag{9}$$

Here every root has $a \neq b$, and each root occurring in these finitely many relations is represented by a word in the chosen generators of $\text{St}_n(R)$.

Proof. Let Q be the group given by this presentation. In the intended semidirect product, p_a translates by $1 + J$ in row a , and $[x_{ab}(r), p_b]$ translates by $r + J$ in that row. We recover these translations and their elementary action from the finite relations, working inside Q until the final inverse-homomorphism argument.

Coefficients. Relation (7) holds for every $s \in R$. To see this, let C be the set of coefficients for which it holds for every permitted triple of indices. By (3), C is an additive subgroup containing F . If $r, s \in C$, choose $h \notin \{a, b, c\}$. There is such an index because $n \geq 4$. Then

$$x_{ab}(rs) = [x_{ah}(r), x_{hb}(s)],$$

and both factors commute with p_c . Thus C is closed under multiplication, so $C = R$.

Independence of a column. For $a \neq b$, define

$$w_a^b(r) = [x_{ab}(r), p_b].$$

If h differs from a, b , then $w_a^b(r)$ commutes with p_h : both $x_{ab}(r)$ and p_b do. Put

$$X = x_{ah}(r), \quad Y = x_{hb}(1), \quad Z = x_{ab}(r).$$

The Steinberg relation $[X, Y] = Z$ gives $XY = ZYX$. Conjugating p_b by these equal products yields

$$\begin{aligned} (XY)p_b(XY)^{-1} &= w_a^h(r)p_hp_b, \\ (ZYX)p_b(ZYX)^{-1} &= p_hw_a^b(r)p_b. \end{aligned}$$

Here X fixes p_b , Y sends p_b to p_hp_b , and Z fixes p_h . Cancelling p_b and using the preceding commutation with p_h proves $w_a^h(r) = w_a^b(r)$. We write this common value as $w_a(r)$.

Commutations. The element $w_a(r)$ commutes with every p_c : choose its defining column $b \notin \{a, c\}$, so that both defining factors commute with p_c . It also commutes with $x_{cd}(s)$ whenever $d \neq a$. For this, choose the defining column $b \notin \{a, c, d\}$. Then $x_{ab}(r)$ commutes with $x_{cd}(s)$ by (5), and p_b commutes with $x_{cd}(s)$ because $b \neq d$.

These facts imply that all the w 's commute. Given $w_a(r)$ and $w_c(s)$, choose $b \notin \{a, c\}$ and write $w_c(s) = [x_{cb}(s), p_b]$. The element $w_a(r)$ commutes with both factors, hence with their commutator. This argument includes $a = c$.

Additivity. Using $[uv, p] = u[v, p]u^{-1}[u, p]$ and any $b \neq a$, we obtain

$$\begin{aligned} w_a(r + s) &= [x_{ab}(r)x_{ab}(s), p_b] \\ &= x_{ab}(r)w_a(s)x_{ab}(r)^{-1}w_a(r) \\ &= w_a(r)w_a(s). \end{aligned}$$

The last equality uses the commutations just proved. Thus $w_a : (R, +) \rightarrow Q$ is a homomorphism. Moreover, $w_a(1) = p_a$ by (8). In particular, any additive torsion of R is already imposed on these translations; in characteristic two, $p_a^2 = w_a(1 + 1) = 1$.

Conjugation. It remains to determine the action of roots whose column is a . For $c \neq a$, choose $b \notin \{a, c\}$ and put

$$X = x_{ca}(s), \quad Y = x_{ab}(r), \quad Z = x_{cb}(sr).$$

The element X fixes p_b , and $XYX^{-1} = ZY$. Consequently

$$Xw_a(r)X^{-1} = [ZY, p_b] = Zw_a(r)Z^{-1}w_c(sr) = w_a(r)w_c(sr). \quad (10)$$

The last equality holds because the column b of Z differs from a . Together with the commutations already proved, (10) is precisely the elementary action on an additive column over the left module R .

The module relations. Relation (9) says $w_a(d) = 1$ for every a and $d \in D$. To kill a left multiple in a prescribed row c , choose $a \neq c$ and conjugate $w_a(d) = 1$ by $x_{ca}(s)$. Equation (10) gives $w_c(sd) = 1$. Additivity now proves that every w_c vanishes on J .

Inverse homomorphisms. Let $P_0 = (R/J)^n \rtimes \text{St}_n(R)$. Sending p_a to translation by $1 + J$ in row a and using the identity on $\text{St}_n(R)$ satisfies all the defining relations of Q , and hence gives a homomorphism $f : Q \rightarrow P_0$. Conversely, the commuting additive maps

$$r + J \longmapsto w_a(r)$$

define a homomorphism $(R/J)^n \rightarrow Q$. Equation (10) and the root commutations give the required equivariance, so this map and the canonical map $\text{St}_n(R) \rightarrow Q$ induce $g : P_0 \rightarrow Q$. The composite gf fixes $\text{St}_n(R)$ and each $p_a = w_a(1)$, so it is the identity on Q . The composite fg fixes $\text{St}_n(R)$ and every translation: in P_0 , the commutator of $x_{ab}(r)$ with translation by $1 + J$ in row b is translation by $r + J$ in row a . Thus f and g are inverse isomorphisms. $\square$

Proposition 3.3. *For the algebra B in Theorem 2.2, the group*

$$\Gamma_0 = B^6 \rtimes \text{St}_6(R), \quad R = B \otimes_k B^{\text{op}},$$

is finitely presented.

Proof. The ring R is finitely presented, so the Krstić–McCool theorem applies to $\text{St}_6(R)$. Lemma 3.1 identifies B with a quotient of R by a finitely generated left ideal. Lemma 3.2, with $n = 6$, gives the result. $\square$

3.3 The transition endomorphism

Write $\text{St} = \text{St}_6(R)$, $E = E_6(R)$, and $\pi = \pi_6 : \text{St} \rightarrow E$. Recall the injective algebra map $\phi : B \rightarrow B$ from Theorem 2.2, and set $\psi = \phi \otimes \phi^{\text{op}} : R \rightarrow R$. This map is injective because tensoring vector spaces over k preserves injections. Both maps may be nonunital.

Lemma 3.4. *The formulas*

$$\theta(x_{ab}(r)) = x_{ab}(\psi(r)), \quad \Theta(m, h) = (\phi^{(6)}(m), \theta(h))$$

define endomorphisms of St and Γ_0 , respectively. Here $\phi^{(6)}$ acts componentwise on B^6 . The induced map on elementary matrices is the injective homomorphism

$$F : E \longrightarrow E, \quad F(g) = I + \psi^{(6)}(g - I), \tag{11}$$

where $\psi^{(6)}$ means entrywise application on 6×6 matrices. Thus $\pi\theta = F\pi$.

Proof. Additivity and multiplicativity of ψ preserve all the Steinberg relations, proving that θ is a homomorphism. For an elementary tensor $r = a \otimes b^{\text{op}}$ and $m \in B$,

$$\phi(r \cdot m) = \phi(amb) = \phi(a)\phi(m)\phi(b) = \psi(r) \cdot \phi(m).$$

By additivity this holds for every $r \in R$. Consequently, for each elementary root,

$$\phi^{(6)}((I + re_{ab})m) = (I + \psi(r)e_{ab})\phi^{(6)}(m).$$

The roots generate St , so this equivariance proves that Θ is an endomorphism of the semidirect product.

For the matrix assertion, write $g = I + A$ and $h = I + C$. Entrywise application of ψ is additive and multiplicative on matrix rings, even though it does not preserve their identity. Since $gh - I = A + C + AC$, we have

$$\begin{aligned} F(gh) &= I + \psi^{(6)}(A) + \psi^{(6)}(C) + \psi^{(6)}(A)\psi^{(6)}(C) \\ &= F(g)F(h). \end{aligned}$$

Also $F(I) = I$, and the same calculation shows $F(g^{-1}) = F(g)^{-1}$. On elementary matrices, $F(I + re_{ab}) = I + \psi(r)e_{ab}$, proving both that F maps E into E and that $\pi\theta = F\pi$. Finally, $F(g) = I$ forces every entry of $g - I$ to vanish because ψ is injective. Thus F is injective. $\square$

We have obtained a finite presentation and compatible transition maps. The map Θ need not be injective: the remaining obstruction lies in the kernel of π . The next section proves that θ annihilates this entire kernel.

4 Annihilating the Steinberg kernel

The affine group of Section 3 uses a Steinberg group, whose matrix action can have a kernel. We now show that the coefficient endomorphism annihilates this entire kernel. The argument uses the splitting pairs from Theorem 2.2, whose elements commute with the coefficient image. It is convenient to state it for an arbitrary unital associative ring.

Recall that $\text{St}_6(R)$ has generators $x_{ij}(a)$, where $1 \leq i \neq j \leq 6$ and $a \in R$, with relations

$$x_{ij}(a)x_{ij}(b) = x_{ij}(a+b), \tag{12}$$

$$[x_{ij}(a), x_{jk}(b)] = x_{ik}(ab) \quad (i, j, k \text{ distinct}), \tag{13}$$

$$[x_{ij}(a), x_{kl}(b)] = 1 \quad (i \neq l, j \neq k). \tag{14}$$

Here $[u, v] = uvu^{-1}v^{-1}$. Let $\pi : \text{St}_6(R) \rightarrow E_6(R)$ send $x_{ij}(a)$ to $I + ae_{ij}$.

Theorem 4.1. *Let R be a unital associative ring, let $\psi : R \rightarrow R$ be an additive multiplicative map, and put $P = \psi(1)$. Suppose that $U_0, U_1, V_0, V_1 \in \text{PRP}$ commute with $\psi(R)$ and satisfy*

$$V_i U_j = \delta_{ij} P, \quad U_0 V_0 + U_1 V_1 = P.$$

Then the homomorphism

$$\theta : \text{St}_6(R) \longrightarrow \text{St}_6(R), \quad x_{ij}(a) \longmapsto x_{ij}(\psi(a)),$$

satisfies $\theta(\ker \pi) = 1$.

For the binary Leavitt algebra $L = L_{\mathbb{F}_2}(1, 2)$, Khanh proves that the natural map $\text{St}_r(L) \rightarrow \text{GL}_r(L)$ is an isomorphism for $r \geq 3$ [11, Theorem 5.4]. Theorem 4.1 concerns an arbitrary ring and annihilation of its matrix kernel under the specified coefficient map.

Additivity and multiplicativity make θ a homomorphism by the defining relations. The map ψ need not preserve the identity. For any additive multiplicative map $\lambda : R \rightarrow R$, the corresponding image of a root word with matrix g has matrix

$$I + \lambda^{(6)}(g - I), \tag{15}$$

where $\lambda^{(6)}$ acts entrywise. This follows on generators and then on products by expanding $gh - I = (g - I) + (h - I) + (g - I)(h - I)$. In particular, coefficient maps preserve the condition of

having identity matrix. We shall use this for maps whose images lie in smaller idempotent corners of R .

The proof has two parts. First, a word with identity matrix becomes central when all its coefficients are placed in a suitable corner. Second, two orthogonal copies of that corner can be interchanged and split. Four corners suffice to make the resulting central element equal to its square.

4.1 Conjugation with an unused index

The use of an index outside the support of a root word is the classical Steinberg centrality argument; see [19, proof of Theorem III.5.2.1]. We record its row-and-column form to justify the corner conjugations inside the Steinberg group. The kernel-annihilation theorem also uses the corner comparisons and four-corner splitting proved below.

Lemma 4.2. *Let $J \subseteq \{1, \dots, 6\}$ and $k \notin J$. For a column c and a row r indexed by J , put*

$$C_k(c) = \prod_{i \in J} x_{ik}(c_i), \quad R_k(r) = \prod_{i \in J} x_{ki}(r_i).$$

If w is a root word using only indices in J , and A is its matrix on those indices, then

$$wC_k(c)w^{-1} = C_k(Ac), \quad wR_k(r)w^{-1} = R_k(rA^{-1}). \quad (16)$$

Consequently, an element represented by a root word omitting index 6 and having identity matrix is central in $\text{St}_6(R)$.

Proof. The factors in each defining product commute. It suffices to check conjugation by $x_{ij}(a)$ with $i, j \in J$. Its only changed column factor and row factor are, respectively,

$$\begin{aligned} x_{ij}(a)x_{jk}(c_j)x_{ij}(a)^{-1} &= x_{ik}(ac_j)x_{jk}(c_j), \\ x_{ij}(a)x_{ki}(r_i)x_{ij}(a)^{-1} &= x_{kj}(-r_ia)x_{ki}(r_i). \end{aligned}$$

The first identity is (13). For the second, reverse the relation $[x_{ki}(r_i), x_{ij}(a)] = x_{kj}(r_ia)$. These operations are multiplication of a column by $I + ae_{ij}$ and of a row by $I - ae_{ij}$. Composition proves (16) inside the Steinberg group.

If w omits 6 and has identity matrix, it therefore commutes with every $x_{i6}(a)$ and $x_{6i}(a)$. It commutes with every remaining root as well, since

$$x_{ij}(a) = [x_{i6}(a), x_{6j}(1)] \quad (i, j \neq 6).$$

These roots generate the group. □

Lemma 4.3. *Let $E, F \in R$ be idempotents with $EF = FE = 0$. Suppose $d, d^* \in R$ satisfy*

$$d = FdE, \quad d^* = Ed^*F, \quad d^*d = E, \quad dd^* = F. \quad (17)$$

Every root word whose coefficients lie in ERE and whose matrix is the identity represents a central element of $\text{St}_6(R)$.

Proof. Here d^* denotes the specified element in (17); no involution on R is assumed. Put

$$W = x_{16}(d)x_{61}(-d^*)x_{16}(d).$$

Its matrix and inverse on the ordered pair $(1, 6)$ are

$$H = \begin{pmatrix} 1-F & d \\ -d^* & 1-E \end{pmatrix}, \quad H^{-1} = \begin{pmatrix} 1-F & -d \\ d^* & 1-E \end{pmatrix}. \quad (18)$$

These formulas follow by multiplying the three elementary matrices and using (17).

Let $y = EyE$ and $k \notin \{1, 6\}$. Applying Lemma 4.2 to the actual word W gives

$$\begin{aligned} Wx_{1k}(y)W^{-1} &= x_{1k}(y), & Wx_{6k}(y)W^{-1} &= x_{1k}(dy), \\ Wx_{k1}(y)W^{-1} &= x_{k1}(y), & Wx_{k6}(y)W^{-1} &= x_{k1}(yd^*). \end{aligned} \quad (19)$$

Indeed $d^*y = yd = 0$, since E and F are orthogonal, and $(1-E)y = y(1-E) = 0$. Roots with both indices outside $\{1, 6\}$ commute with W .

For the two roots internal to this pair, first write

$$x_{16}(y) = [x_{1k}(y), x_{k6}(E)], \quad x_{61}(y) = [x_{6k}(y), x_{k1}(E)].$$

Their conjugates under W are, by (19),

$$[x_{1k}(y), x_{k1}(d^*)], \quad [x_{1k}(dy), x_{k1}(E)],$$

respectively. Both are words omitting index 6; no relation for opposite roots is needed here. Thus W conjugates every word in the statement to a word omitting 6. Its matrix remains the identity, so Lemma 4.2 makes it central. The original element is central as well. $\square$

4.2 Interchanging and splitting corners

For a unit $a \in R^\times$, define

$$w_{ij}(a) = x_{ij}(a)x_{ji}(-a^{-1})x_{ij}(a), \quad h_{ij}(a) = w_{ij}(a)w_{ij}(-1).$$

Direct multiplication gives

$$\begin{aligned} \pi(w_{ij}(a))|_{\{i,j\}} &= \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix}, \\ \pi(h_{ij}(a))|_{\{i,j\}} &= \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}; \end{aligned} \quad (20)$$

both matrices are the identity on the other indices.

Lemma 4.4. *Let $\psi : R \rightarrow R$ be an additive multiplicative map. Let E, F, d, d^* satisfy (17), with E, F orthogonal idempotents, and suppose all four elements commute with $\psi(R)$. For $z \in \ker \pi$, denote its images under the coefficient maps $a \mapsto \psi(a)E$ and $a \mapsto \psi(a)F$ by z_E and z_F . Then $z_E = z_F$, and these elements are central.*

Proof. The two coefficient maps are multiplicative because E and F are idempotent and commute with $\psi(R)$. Their root words have identity matrix by (15), so Lemma 4.3, in both orders, gives centrality.

Put $b = d + d^* + 1 - E - F$. The squares of d and d^* vanish, their cross products are F and E , and $1 - E - F$ is an idempotent annihilating them on both sides. Thus

$$b^2 = 1, \quad bEb^{-1} = F,$$

and b commutes with $\psi(R)$. Consider the element

$$D = h_{12}(b)h_{34}(b)h_{56}(b) \in \text{St}_6(R).$$

Its matrix is diagonal with b in all six positions. More particularly, we claim the exact Steinberg identity

$$Dx_{ij}(y)D^{-1} = x_{ij}(byb^{-1}) \quad (y \in R). \quad (21)$$

If i and j belong to different pairs among $(1, 2), (3, 4), (5, 6)$, apply Lemma 4.2 to the two relevant factors of D . The pair containing i multiplies the coefficient on the left by b , and the pair containing j multiplies it on the right by b^{-1} ; the third factor commutes with the root. Here $b = b^{-1}$ makes the action the same at either position of a pair. If i, j belong to the same pair, choose k in another pair and use

$$x_{ij}(y) = [x_{ik}(y), x_{kj}(1)].$$

The cross-pair case conjugates its factors to $x_{ik}(byb^{-1})$ and $x_{kj}(1)$, proving (21) also in this case.

Since b commutes with $\psi(R)$ and interchanges E, F , (21) implies $Dz_ED^{-1} = z_F$. Centrality gives $z_E = z_F$. $\square$

Lemma 4.5. *If $E, F \in R$ are orthogonal idempotents, then the subgroup of $\text{St}_6(R)$ generated by roots with coefficients in ERE commutes with the subgroup generated by roots with coefficients in FRF .*

Proof. For $a \in ERE$ and $c \in FRF$, one has $ac = ca = 0$. Roots with these coefficients commute by the defining relations unless their indices are opposite: chain commutators have coefficient ac or $-ca$, and roots in the same position commute by additivity. For the remaining pair $x_{ij}(a), x_{ji}(c)$ choose k distinct from i, j and write

$$x_{ij}(a) = [x_{ik}(a), x_{kj}(E)].$$

The root $x_{ji}(c)$ commutes with the first factor because $ca = 0$, and with the second because $Ec = 0$. It therefore commutes with their commutator. $\square$

We can now finish the kernel calculation. The preceding lemmas allow us to compare corner images as group elements, and to split an image over an orthogonal sum without a residual central factor.

Proof of Theorem 4.1. Define four idempotents by

$$E_i = U_i V_i \quad (i = 0, 1), \quad E_{0i} = U_0 U_i V_i V_0 \quad (i = 0, 1).$$

They commute with $\psi(R)$, and the pair relations give the orthogonal decompositions

$$P = E_0 + E_1, \quad E_0 = E_{00} + E_{01}. \quad (22)$$

Each of E_0, E_{00}, E_{01} is orthogonal to E_1 and equivalent to it as in (17). To check this explicitly, put

$$\begin{aligned} U_{00} &= U_0 U_0, & V_{00} &= V_0 V_0, \\ U_{01} &= U_0 U_1, & V_{01} &= V_1 V_0. \end{aligned}$$

and, for $v \in \{0, 00, 01\}$, set

$$d_v = U_1 V_v, \quad d_v^* = U_v V_1.$$

The relations $V_v U_v = P$ and $U_v V_v = E_v$ imply

$$d_v = E_1 d_v E_v, \quad d_v^* = E_v d_v^* E_1, \quad d_v^* d_v = E_v, \quad d_v d_v^* = E_1.$$

All these elements commute with $\psi(R)$.

Figure 1 separates the two orthogonal decompositions from the equivalences between their summands. Comparing each of E_0, E_{00}, E_{01} with E_1 will identify their images of a kernel element.

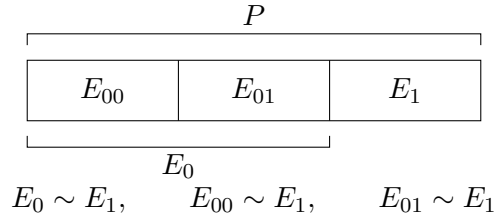


Figure 1: The decompositions $P = E_0 + E_1$ and $E_0 = E_{00} + E_{01}$, shown schematically. The three disjoint cells are orthogonal; the bracketed corner E_0 contains its two children. Here $\sim$ denotes the equivalence in (17).

Fix $z \in \ker \pi$ and let z_Q denote its image under the coefficient map $a \mapsto \psi(a)Q$ for each of these idempotents. Lemma 4.4, applied three times with the second corner E_1 , gives

$$z_{E_0} = z_{E_1} = z_{E_{00}} = z_{E_{01}}. \quad (23)$$

By additivity, the second split in (22) splits every root image into its two child images. Lemma 4.5 permits their factors to be collected separately in any word. Hence

$$z_{E_0} = z_{E_{00}} z_{E_{01}} = z_{E_0}^2,$$

so $z_{E_0} = 1$ and then $z_{E_1} = 1$. Finally $\psi(a)P = \psi(a)$. The first split in (22), with the same commutation argument, therefore gives

$$\theta(z) = z_P = z_{E_0} z_{E_1} = 1.$$

□

For $R = B \otimes_k B^{\text{op}}$ and $\psi = \phi \otimes \phi^{\text{op}}$, the elements P, U_i, V_i constructed in Section 2 satisfy the support and commutation hypotheses and the pair identities (1). Theorem 4.1 therefore applies to the Steinberg transition of Section 3: it annihilates the matrix kernel in one step.

5 The two-transitive envelope

We now turn the algebraic construction into the required permutation representation. Proposition 3.3 and Lemma 3.4 give the finitely presented group

$$\Gamma_0 = B^6 \rtimes \text{St}_6(R)$$

and its endomorphism $\Theta = (\phi^{(6)}, \theta)$. Theorem 4.1 says that θ kills the kernel of the matrix map. A mapping torus will remove this kernel while retaining finite presentation. We then use the simple subalgebra $A \subseteq B$ to obtain two-transitivity.

5.1 The mapping torus and its linear part

We first record explicitly why a mapping torus has the needed description even when its defining endomorphism is not injective.

Lemma 5.1. *Let Q be a group and $f : Q \rightarrow Q$ an endomorphism. Write $L = \varinjlim (Q, f)$, with classes $[q, j]$ for $j \geq 0$ and the convention $[q, j] = [f(q), j + 1]$. The map*

$$\alpha([q, j]) = [q, j + 1]$$

is an automorphism of L , and

$$\langle Q, t \mid t^{-1}qt = f(q) \ (q \in Q) \rangle \cong L \rtimes_{\alpha} \mathbb{Z}.$$

If Q is finitely presented, so is this group.

Proof. Equality in the direct limit means equality after sufficiently many forward transitions. Thus α is well defined, as is $\beta([q, j]) = [f(q), j]$. The defining equivalence gives $\alpha\beta = \beta\alpha = \text{id}$. Let t act on L by α . Then

$$t^{-1}[q, 0]t = \beta([q, 0]) = [f(q), 0],$$

so the displayed presentation maps to $L \rtimes_{\alpha} \mathbb{Z}$. In the other direction, send

$$[q, j] \mapsto t^j q t^{-j}.$$

The relation $t^{-1}qt = f(q)$ implies $t^{j+1}f(q)t^{-(j+1)} = t^j q t^{-j}$, proving that this map is well defined. It intertwines α with conjugation by t , and hence extends to a homomorphism from the semidirect product. The two maps are inverse on Q and t .

For a finite presentation $Q = \langle X \mid \mathcal{R} \rangle$, choose a word representing $f(x)$ for each $x \in X$. Adjoin t and the finitely many relations $t^{-1}xt = f(x)$ for $x \in X$. These imply the required relations for every element of Q , since both conjugation and f are homomorphisms. $\square$

Apply Lemma 5.1 to (Γ_0, Θ) and denote the resulting finitely presented group by Γ . Put

$$M_{\infty} = \varinjlim ((B, +), \phi), \quad E_{\infty} = \varinjlim (E_6(R), F), \quad F(g) = I + \psi^{(6)}(g - I).$$

The transition F is the injective homomorphism of Lemma 3.4; recall the matrix map $\pi : \text{St}_6(R) \rightarrow E_6(R)$.

Lemma 5.2. *There are compatible isomorphisms*

$$\varinjlim(\Gamma_0, \Theta) \cong M_\infty^6 \rtimes E_\infty, \quad \Gamma \cong (M_\infty^6 \rtimes E_\infty) \rtimes \mathbb{Z}.$$

In the second isomorphism, conjugation by t shifts the stage index forward by one on both M_∞ and E_∞ .

Proof. The square $\pi\theta = F\pi$ induces a surjection from the Steinberg direct limit to E_∞ . If a class represented by $h \in \text{St}_6(R)$ maps to the identity, some iterate of F sends $\pi(h)$ to I . Since F is injective, $\pi(h) = I$ already. Theorem 4.1 now gives $\theta(h) = 1$, so that class is trivial.

The compatibility of the module action is

$$\phi^{(6)}(gv) = F(g)\phi^{(6)}(v) \quad (g \in E_6(R), v \in B^6). \quad (24)$$

Indeed $\phi(r \cdot b) = \psi(r) \cdot \phi(b)$ for $r \in R$ and $b \in B$, and separating g into $I + (g - I)$ gives exactly the modified-identity formula for F . Thus matrices and vectors act on one another after passage to any common stage.

The stage maps

$$(v, h) \mapsto ([v, j], [\pi(h), j])$$

therefore induce a homomorphism from $\varinjlim(\Gamma_0, \Theta)$ to $M_\infty^6 \rtimes E_\infty$. It is surjective: choose a common stage for a vector and a matrix, and then choose a Steinberg preimage of that matrix. If a representative (v, h) maps to the identity, injectivity of ϕ gives $v = 0$, and injectivity of F gives $\pi(h) = I$. Consequently $\Theta(v, h) = (0, 1)$ by Theorem 4.1. The homomorphism is injective. The construction respects stage shifts, so Lemma 5.1 gives the second isomorphism as well. $\square$

5.2 A faithful action and two-transitivity

Set $X = M_\infty^6$. The subgroup M_∞^6 acts on X by translations, and E_∞ acts by matrices after passing to a common stage, as in (24). Let t act by the forward stage shift. This shift intertwines the matrix actions, so these formulas define an action of Γ on X .

Proposition 5.3. *The set X is countably infinite. The action of Γ on X is faithful and two-transitive, and all its point stabilizers are finitely generated.*

Proof. The stage filtration. Since ϕ is injective, identify the j th additive stage with its image $M_j \subseteq M_\infty$, and put $X_j = M_j^6$. The stages are nested because $[b, j] = [\phi(b), j+1]$. They are strictly nested: every $\phi(b)$ satisfies $p\phi(b) = \phi(b)$, where $p = \phi(1_B) \neq 1_B$, so 1_B cannot belong to $\phi(B)$. Thus $[1_B, j+1] \notin M_j$, and

$$X_0 \subsetneq X_1 \subsetneq X_2 \subsetneq \cdots. \quad (25)$$

The algebra B is countable, being a quotient of the free associative algebra over the finite field k on finitely many generators. Consequently $X = \bigcup_{j \geq 0} X_j$ is countable; the strict inclusions make it infinite.

Faithfulness of the affine part. Take a nonidentity element of E_∞ , represented by a matrix $g \neq I$ at stage j . Choose a nonzero entry r of $g - I$, say in row a and column b . At stage $j+1$, let v have b_* in coordinate b and zero in the other coordinates. The a th coordinate of $F(g)v - v$ is

$$\psi(r) \cdot b_* \neq 0$$

by Theorem 2.2. It remains nonzero in the direct limit because ϕ is injective. This proves faithfulness of the linear action. If an affine element acts identically, evaluating it at zero first makes its translation zero; linear faithfulness then makes the element trivial.

Faithfulness of the whole group. Every affine limit element and its inverse have representatives at some common stage. At every later stage their translation vectors, matrices, and inverse matrices act within that stage. Hence, for each affine limit element a ,

$$a(X_j) = X_j \quad \text{for all sufficiently large } j. \quad (26)$$

In contrast,

$$t^k(X_j) = X_{j+k} \quad \text{whenever } j \geq 0 \text{ and } j+k \geq 0.$$

For $k \neq 0$, this differs from X_j by (25). Thus no nonzero power of t agrees, as a permutation, with an affine limit element. If at^k acts trivially, then t^k acts as a^{-1} , forcing $k = 0$ by (26). Affine faithfulness then gives $a = 1$.

Two-transitivity. We show that E_∞ is transitive on the nonzero vectors of X . Given two such vectors, choose a common representing stage and advance once more. All their coordinates now lie in A , since $\phi(B) \subseteq A$, and the vectors remain nonzero. We work in this one stage, with the same simple algebra A and its identity e .

If a vector has a nonzero coordinate $m_j \in A$, simplicity gives $Am_jA = A$: the finite sums of elements am_jb form a nonzero two-sided ideal. Thus, for any $d \in A$, there are finitely many $a_\nu, b_\nu \in A$ with

$$d = \sum_{\nu} a_{\nu} m_j b_{\nu}.$$

The coefficient $r = \sum_{\nu} a_{\nu} \otimes b_{\nu}^{\text{op}} \in R$ then satisfies $r \cdot m_j = d$. For $l \neq j$, an elementary matrix in row l , column j , with coefficient r adds d to coordinate l and changes no other coordinate. It can therefore set that coordinate to any prescribed value in A .

If $j \neq 6$, set the last coordinate to e . If $j = 6$, first set coordinate 1 to e , then use it to set the last coordinate to e . Use the last coordinate to kill all the others. This sends the vector to $(0, 0, 0, 0, 0, e)$. Applying the procedure to both original vectors sends them to the same vector at the same stage, proving the asserted linear transitivity. Translations, followed by these linear maps and another translation, take any ordered pair of distinct points to any other such pair. The action is two-transitive.

Point stabilizers. Both E_∞ and t fix zero, so the semidirect-product description gives

$$\text{Stab}_{\Gamma}(0) = E_\infty \rtimes \langle t \rangle \cong \Gamma/X.$$

This is a finitely generated group, since Γ is finitely presented. More explicitly, t and the images of a finite generating set of the base Steinberg group generate it: conjugating by positive powers of t reaches every linear stage. All other point stabilizers are conjugate to this one, since translations act transitively. $\square$

5.3 Embedding the input group and finishing the proof

The last step is to put the original group into the elementary matrix part of Γ . Here the commutator property of Lemma 2.1 is essential.

Proof of Theorem 1.2. Let $\widehat{G}$ be the auxiliary group of Lemma 2.1. Theorem 2.2 supplies an injective homomorphism $\iota : \widehat{G} \rightarrow A^\times$, where units in A have identity e . Define

$$\nu(h) = 1_B - e + \iota(h) \quad (h \in \widehat{G}).$$

The element $1_B - e$ annihilates A on both sides. Consequently

$$\nu(h)\nu(h') = \nu(hh'), \quad \nu(h)^{-1} = \nu(h^{-1}).$$

Injectivity of ι proves injectivity of ν . Composing with the injective unital map $B \rightarrow R$, $b \mapsto b \otimes 1_B$, gives a group embedding $\bar{\nu} : \hat{G} \rightarrow R^\times$. The tensor map is injective because B is a nonzero vector space over k .

For $u \in R^\times$, let $D_{ij}(u)$ be the diagonal matrix with entries u, u^{-1} in positions i, j and 1 elsewhere. Equation (20) shows that $D_{ij}(u) \in E_6(R)$. Let s be the stable letter in $\hat{G}$ and set

$$a_g = \bar{\nu}((g, g)), \quad b = \bar{\nu}(s).$$

By Lemma 2.1, $[a_g, b] = \bar{\nu}((g, 1))$. Taking the commutator of balanced diagonal matrices on two different pairs of coordinates gives

$$[D_{12}(a_g), D_{13}(b)] = \text{diag}(\bar{\nu}((g, 1)), 1, 1, 1, 1, 1).$$

Thus the injective homomorphism given by the diagonal on the right has image in $E_6(R)$. Since the transition F is injective, this embeds G in E_∞ , and hence in Γ by Lemma 5.2. Proposition 5.3 and the finite presentation of Γ give all the remaining assertions of Theorem 1.2. $\square$

Proof of Theorem 1.1. Suppose first that G is finitely generated and has decidable word problem. Theorem 1.2 embeds it in a finitely presented group Γ with a faithful two-transitive action on a countably infinite set and finitely generated point stabilizers. There is exactly one orbit on unordered two-element subsets. These are all the hypotheses of Theorem 2.3, which embeds Γ in a finitely presented simple group S . The group S is nontrivial, since Γ acts transitively on an infinite set. This proves the required direction, including trivial and finite input groups.

For the converse, we give the classical decision argument attributed by Boone and Higman to Kuznetsov [6, p. 52]. Let $S = \langle Y \mid \mathcal{R} \rangle$ be a nontrivial finitely presented simple group. On an input word w , run the following two searches in parallel:

- (i) enumerate presentation proofs that $w = 1$ follows from $\mathcal{R}$;
- (ii) enumerate, for every $y \in Y$, a presentation proof that $y = 1$ follows from $\mathcal{R} \cup \{w\}$, stopping when a proof has been found for every generator.

Consequences of a finite group presentation are recursively enumerable, by enumerating finite products of conjugates of relators and their inverses. If $w = 1$ in S , the first search succeeds, whereas the second cannot succeed for every generator because S is nontrivial. If $w \neq 1$, its normal closure in the simple group S is all of S ; thus the second search succeeds for each of the finitely many generators, whereas the first never succeeds. Exactly one search terminates, deciding the word problem in S .

For a finitely generated subgroup of S , fix words in Y representing its generators and substitute these words into each input. This transfers the decider to that subgroup. Finally, the same substitution argument between any two finite generating sets of a group proves that decidability of its word problem is independent of the chosen finite generating set. $\square$

6 The algebra construction

We now prove Theorem 2.2. The construction has two parts. A pair of monoids supplies a simple algebra and an embedding into a finitely presented algebra. Partial affine maps then give a corner with two equivalent orthogonal subcorners. A separator in the monoid construction will prove the tensor detection property.

6.1 The monoid input

We first define the auxiliary monoid used in the construction. Let $\mathcal{P}$ be the monoid of partial maps of $\mathbb{Q}$ generated by the full affine maps

$$u(x) = x/2, \quad v(x) = (x+1)/2,$$

their inverses, and the partial identities e_0, e_1 on $[0, \infty) \cap \mathbb{Q}$ and $[1, \infty) \cap \mathbb{Q}$, respectively. Composition applies the right-hand factor first. The identity $1_{\mathcal{P}}$ is the full identity map.

Lemma 6.1. *The monoid $\mathcal{P}$ has decidable word problem. Each of its elements is a map $x \mapsto ax+b$ on a nonempty rational tail $[r, \infty) \cap \mathbb{Q}$, or on all of $\mathbb{Q}$, where $a = 2^m$ for some $m \in \mathbb{Z}$, and b, r are dyadic rationals.*

Proof. Represent a full domain by $r = -\infty$. The composition of the data (a, b, r) and (c, d, s) is

$$(ac, ad+b, \max\{s, (r-d)/c\}).$$

The generators have such data, and the displayed operation preserves them. Each domain is infinite, so two represented maps agree precisely when their domains, slopes and intercepts agree. The data and their comparison are computable using dyadic arithmetic. $\square$

For a monoid K with zero $\Omega \neq 1_K$, write

$$T = (K \setminus \{\Omega\}) \times \mathcal{P} \sqcup \{0_T\}.$$

The element 0_T is absorbing. Nonzero pairs multiply coordinatewise, except that a product whose K -coordinate is Ω is replaced by 0_T . Its identity is $(1_K, 1_{\mathcal{P}})$.

Theorem 6.2. *Let $\hat{G}$ be a finitely generated group with decidable word problem. There exist a finitely presented monoid K with zero $\Omega \neq 1_K$, and a finitely generated monoid H with decidable word problem and zero $\zeta \neq 1_H$, with the following properties.*

- (i) *There is an injective group homomorphism $\hat{G} \longrightarrow H^\times$.*
- (ii) *For the monoid T defined above, there is an injective unital monoid homomorphism $\kappa : T \longrightarrow H$ with $\kappa(0_T) = \zeta$.*
- (iii) *For every finite ordered tuple $h_1, \dots, h_d$ of distinct nonzero elements of H , with $d \geq 1$, there are $L, R \in H$ such that*

$$Lh_1R = 1_H, \quad Lh_iR = \zeta \quad (2 \leq i \leq d).$$

- (iv) *There is an injective multiplicative map $c : H \longrightarrow K$ whose image is disjoint from $\{1_K, \Omega\}$. We write $c_h = c(h)$; this map is not required to preserve identities or zeros.*
- (v) *There is an element $z \in H$ such that the map*

$$(\tau, \tau') \longmapsto \kappa(\tau)z\kappa(\tau')$$

is an injection from $(T \setminus \{0_T\})^2$ into $H \setminus \{\zeta\}$.

The proof of Theorem 6.2 occupies Section 7, using the finite compiler proved in Appendix A. In the construction, ζ is the zero symbol and z is the class of a distinct separator letter **z**.

6.2 A simple algebra inside a finitely presented algebra

Throughout this section $k = \mathbb{F}_2$. For any monoid D with zero $0_D \neq 1_D$, its contracted monoid algebra $k_0[D]$ has k -basis $D \setminus \{0_D\}$. Basis elements multiply as in D , with a product equal to 0_D interpreted as the zero vector. It is a unital algebra, with identity the basis element 1_D .

To prove Theorem 2.2 for a given group L , apply Theorem 6.2 with $\widehat{G} = L$ and set

$$A_0 = k_0[H], \quad B = k_0[K].$$

Both algebras are nonzero. We distinguish the monoid elements ζ, Ω from algebra zero, and distinguish c_ζ from all three: it is a nonzero basis element of B .

The monoid requirements now serve distinct purposes. The selectors in Theorem 6.2(iii) isolate one coefficient of a nonzero element of A_0 , forcing simplicity. The maps in (ii) and (iv) provide algebra embeddings in opposite directions, but treat zero differently: κ preserves it, whereas the linear extension of c requires a correction by c_ζ . Finally, the separator in (v) keeps the ordered pairs indexing a tensor basis distinct, which will supply the tensor detector.

Lemma 6.3. *The algebra A_0 is simple, the algebra B is finitely presented, and there are injective algebra homomorphisms*

$$i : A_0 \longrightarrow B, \quad \rho : B \otimes_k k[\mathcal{P}] \longrightarrow A_0.$$

The map ρ is unital. The map i is nonunital and has support identity $e = i(1_{A_0}) \neq 1_B$.

Proof. Let $a = \sum_{j=1}^d \lambda_j h_j$ be a nonzero element of A_0 , with distinct nonzero support elements h_j and nonzero coefficients $\lambda_j \in k$. Choose the selectors from Theorem 6.2(iii). In the contracted algebra they give $LaR = \lambda_1 1_{A_0}$. Hence every nonzero two-sided ideal contains the identity, proving simplicity. This argument also covers a support element equal to 1_H .

The linear extension of c alone would not respect the contraction of the zero class, since $c_\zeta \neq \Omega$. For $h \neq \zeta$, define

$$i(h) = c_h - c_\zeta$$

and extend linearly. Multiplicativity of c gives

$$c_h c_{h'} = c_{hh'}, \quad c_h c_\zeta = c_\zeta c_h = c_\zeta, \quad c_\zeta^2 = c_\zeta.$$

Consequently

$$(c_h - c_\zeta)(c_{h'} - c_\zeta) = c_{hh'} - c_\zeta.$$

When $hh' = \zeta$, this is zero, as required. The vectors c_h are distinct nonzero basis elements of B . In particular, the coefficient of c_h , for $h \neq \zeta$, in $i(a)$ is exactly the coefficient of h in a , proving injectivity. Its support identity is

$$e = c_{1_H} - c_\zeta, \quad e^2 = e, \quad ei(a) = i(a)e = i(a).$$

Neither basis vector in this expression is 1_K ; therefore $e \neq 1_B$, and indeed $1_B \notin i(A_0)$.

A finite presentation for B is obtained from the finite monoid presentation for K . We may adjoin a generator Ω and a relation identifying it with a word representing the zero, so that Ω is a presentation letter. Take the same generators in a free associative unital k -algebra, replace every equation $u = v$ by $u - v = 0$, and impose $\Omega = 0$. To verify this presentation, first omit the last relation. The ideal of binomial relations is precisely the span of all differences of words in the same monoid congruence class. One inclusion follows because each contextual use of a defining equation belongs to the ideal, and a chain of such uses telescopes. The reverse inclusion follows by

mapping each word to its congruence class in the free vector space on K . The quotient thus has basis K ; imposing $\Omega = 0$ removes exactly its absorbing zero class. It gives B . Adding the relation $2 \cdot 1 = 0$ to the same generators and relations also presents B as a unital ring.

Finally, nonzero basis pairs give a unital algebra isomorphism

$$B \otimes_k k[\mathcal{P}] \cong k_0[T].$$

It sends $b \otimes p$, for basis elements $b \neq \Omega$ and $p \in \mathcal{P}$, to (b, p) ; products that acquire K -coordinate Ω vanish on both sides. The linear extension of κ gives the required injective unital map ρ . $\square$

Identify $A = i(A_0) \subseteq B$. It is a simple algebra with its own identity e , and Theorem 6.2(i) embeds L in $A^\times$: a unit of H and its inverse give inverse basis elements in A_0 , which i sends to units relative to e . We next construct an injective nonunital endomorphism of B whose image lies in A .

6.3 Two equivalent subcorners

For a dyadic rational a , let e_a denote the partial identity on $[a, \infty) \cap \mathbb{Q}$. These maps belong to $\mathcal{P}$: $u^{-1}v$ is translation by 1, its powers and conjugates by powers of u give all dyadic translations, and conjugating e_0 by translation by a gives e_a . In the monoid itself,

$$e_a e_b = e_{\max\{a, b\}}, \quad w e_a w^{-1} = e_{w(a)} \quad (27)$$

for every available full increasing affine map w . Thus these are equalities of basis elements in the abstract algebra $k[\mathcal{P}]$.

We require a nonzero corner that decomposes into two orthogonal copies of itself. The identities in the next lemma have the rectangular-inverse form familiar from Leavitt's module-type construction [14, Section 3], with the corner idempotent as identity. Here we realize them directly using the partial affine maps above.

Lemma 6.4. *There is a nonzero idempotent $q \in k[\mathcal{P}]$ and elements $s'_0, s'_1, t'_0, t'_1 \in qk[\mathcal{P}]q$ such that*

$$t'_i s'_j = \delta_{ij} q, \quad s'_0 t'_0 + s'_1 t'_1 = q.$$

Proof. Set $q = e_0 - e_1$. Its two basis elements and $1_{\mathcal{P}}$ are distinct, so $q \neq 0, 1$; Equation (27) gives $q^2 = q$. Put

$$w_0 = u, \quad w_1 = v, \quad q_0 = w_0 q w_0^{-1} = e_0 - e_{1/2}, \quad q_1 = w_1 q w_1^{-1} = e_{1/2} - e_1.$$

Using the same tail identities and expanding in the monoid basis gives

$$q_0^2 = q_0, \quad q_1^2 = q_1, \quad q_0 + q_1 = q, \quad q_0 q_1 = q_1 q_0 = 0.$$

For example, $q_0 q_1 = e_{1/2} - e_1 - e_{1/2} + e_1 = 0$. In particular $q q_j = q_j q = q_j$.

Define $s'_j = w_j q$ and $t'_j = q w_j^{-1}$. Then $s'_j t'_j = q_j$ and $t'_j s'_j = q$. The identities $s'_j = q_j w_j$ and $t'_j = w_j^{-1} q_j$, together with $q q_j = q_j q = q_j$, show

$$q s'_j = s'_j q = s'_j, \quad q t'_j = t'_j q = t'_j.$$

Moreover $t'_j q_j = t'_j$ and $q_j s'_j = s'_j$. For $i \neq j$ we therefore have

$$t'_i s'_j = t'_i q_i q_j s'_j = 0.$$

Finally $\sum_j s'_j t'_j = q_0 + q_1 = q$. All calculations use Equation (27) and linear operations in $k[\mathcal{P}]$. $\square$

Define

$$j : B \longrightarrow A_0, \quad j(b) = \rho(b \otimes q), \quad \phi = i \circ j : B \longrightarrow B.$$

These maps are nonunital algebra homomorphisms. The map j is multiplicative because $q^2 = q$, and it is injective because $q \neq 0$ and tensor products are over the field k . Hence ϕ is injective and $\phi(B) \subseteq A$. Let

$$p = \phi(1_B), \quad s_l = i\rho(1_B \otimes s'_l), \quad t_l = i\rho(1_B \otimes t'_l) \quad (l = 0, 1).$$

The idempotent p belongs to A , so $p \neq 1_B$. Applying the two algebra maps to Lemma 6.4 gives

$$s_l, t_l \in pBp, \quad t_l s_j = \delta_{lj} p, \quad s_0 t_0 + s_1 t_1 = p. \quad (28)$$

These elements commute with $\phi(B)$. Indeed,

$$(b \otimes q)(1_B \otimes s'_l) = b \otimes s'_l = (1_B \otimes s'_l)(b \otimes q),$$

and the identical calculation holds for t'_l ; now apply $i\rho$. Thus the required self-embedding and binary pairs centralizing $\phi(B)$ have been constructed. It remains to verify detection for the tensor action.

6.4 A separating tensor vector

As in Section 3, set $R = B \otimes_k B^{\text{op}}$, acting on B by $(a \otimes b^{\text{op}})m = amb$, and define

$$\psi = \phi \otimes \phi^{\text{op}}.$$

We need one vector on which $\psi(r)$ acts nontrivially for every $r \neq 0$. The separator gives this simultaneously for all r , by realizing distinct basis tensors as distinct monoid elements.

Let $W = \rho(B \otimes_k k[\mathcal{P}]) \subseteq A_0$, with basis $\{\kappa(\tau) : \tau \in T \setminus \{0_T\}\}$. Part (v) of Theorem 6.2 proves that the linear map

$$D : W \otimes_k W^{\text{op}} \longrightarrow A_0, \quad D(x \otimes y^{\text{op}}) = xzy,$$

is injective: it sends distinct basis tensors to distinct nonzero basis elements of A_0 . This includes tensors with an identity factor. Since $j(B) \subseteq W$, the composite

$$R \xrightarrow{j \otimes j^{\text{op}}} W \otimes_k W^{\text{op}} \xrightarrow{D} A_0 \xrightarrow{i} B$$

is a sequence of injective linear maps. For $r = \sum_{\nu} a_{\nu} \otimes b_{\nu}^{\text{op}}$, its value is

$$\begin{aligned} i(D((j \otimes j^{\text{op}})(r))) &= \sum_{\nu} i(j(a_{\nu})) i(z) i(j(b_{\nu})) \\ &= \psi(r) \cdot i(z). \end{aligned}$$

Thus $b_* = i(z)$ satisfies

$$r \neq 0 \implies \psi(r) \cdot b_* \neq 0. \quad (29)$$

The assertion concerns the action after ψ ; its proof uses the explicit separator vector.

Proof of Theorem 2.2. The algebra B is nonzero and finitely presented by Lemma 6.3. Its subalgebra $A = i(A_0)$ is simple with identity e , and contains L in its unit group. The map ϕ is injective, has image in A , and has proper support p . Equation (28) supplies the binary pairs centralizing $\phi(B)$, which give the tensor identities (1) as explained in Section 2. Finally, Equation (29) supplies the required vector b_* . These are all the asserted data. $\square$

7 Coding with a decreasing dependency bound

We prove Theorem 6.2. The two monoids in that theorem must contain information about one another: the finitely presented monoid will multiply representatives of the decidable monoid, while the latter will contain a coded copy of the former. We arrange this using a program whose evaluations on words of length n depend only on its values at strictly smaller lengths. The other ingredient is a family of separated words that supplies the finite-tuple selectors.

7.1 The bounded compiler

Fix a finite alphabet Σ containing the disjoint triples

$$(\mathbf{b}_T, \mathbf{a}_T, \mathbf{c}_T), \quad (\mathbf{b}_G, \mathbf{a}_G, \mathbf{c}_G)$$

and the six further letters $\mathbf{z}, \mathbf{d}, \mathbf{l}, \mathbf{r}, \mathbf{h}, \mathbf{\zeta}$. Fix a total order with $\mathbf{a}_T < \mathbf{c}_T$ and $\mathbf{a}_G < \mathbf{c}_G$. Words are ordered by *shortlex*: first by length and then lexicographically. We use an acceptable programming system for partial functions $\Sigma^* \rightarrow \Sigma^*$, writing f_η for the function with index η . In particular, universal evaluation and effective substitution of parameters are available; integers and tuples are represented by fixed computable word encodings.

The following finite construction is proved in Appendix A. Its conditional statement is essential: the program index determines the presentation without running the program. The evaluator is also fixed independently of a length bound; its termination and compatibility with equations will be proved from the program's behavior within that bound.

Lemma 7.1 (Bounded compiler). *From a program index η one can compute a finite monoid presentation K_η with a zero letter Ω , a finite presentation alphabet X_η , and an additive grade γ on words, such that $0 \leq \gamma(x) \leq M = 2$ for every letter and $\gamma(\Omega) = 0$. There is a partial word algorithm N_η , independent of any budget, with the following properties.*

The alphabet contains letters $\alpha, \mathbf{Q}, \omega$ and right anchor letters a_R for $a \in \Sigma$, all mutually distinct and different from Ω . For a word u , let u_R replace each letter a by a_R , and set

$$\text{raw}(u) = \alpha \mathbf{Q} u_R \omega \quad (u \in \Sigma^*).$$

Unconditionally, the algorithm fixes the empty word, Ω , and every literal word $\text{raw}(u)$, and sends every word containing Ω to Ω . On an input word a , its only calls to f_η are on words of length at most $\gamma(a)$; apart from those calls it performs finite computations.

Suppose that, for some integer $m \geq 0$,

$$f_\eta(w) \text{ is defined and } |f_\eta(w)| \leq |w| \quad (|w| \leq m), \quad (30)$$

$$f_\eta(x f_\eta(w) y) = f_\eta(x w y) \quad (|x w y| \leq m). \quad (31)$$

Then N_η terminates on all words of grade at most m . Each such word is joined to its literal output by a finite derivation in the presentation whose intermediate grades do not exceed the input grade. A single presentation equation, used in any context, preserves the output whenever both endpoints have grade at most m . Consequently

$$N_\eta(u) = N_\eta(v) \implies N_\eta(xuy) = N_\eta(xvy) \quad \text{if } \gamma(xuy), \gamma(xvy) \leq m. \quad (32)$$

If (30)–(31) hold for all m , then N_η decides equality in K_η and

$$N_\eta(\text{raw}(u) \text{raw}(v)) = \text{raw}(f_\eta(uv)). \quad (33)$$

The presentation is computed without evaluating f_η .

For clarity, (32) follows from the preceding derivation statement. Transplant the paths from u and v to their common output into $x(-)y$. Neither contextual path leaves the budget, so one-step invariance identifies their outputs. A general machine encoding without this bounded conclusion would not suffice below. The raw singleton rule has a different role: it will distinguish the elements used for the injection $H \rightarrow K$. Multiplication of two raw words, rather than evaluation of a singleton, will encode the product in H through (33).

7.2 Codes and finite rule tables

First consider any index η . The auxiliary monoid $\mathcal{P}$ has decidable equality by Lemma 6.1.

Let Y be the disjoint union of X_η and a fixed finite generating alphabet for $\mathcal{P}$. For a Y -word v , write $v_K, v_{\mathcal{P}}$ for its projections, obtained by deleting the letters of the other alphabet. Its *tested class* is zero if $N_\eta(v_K) = \Omega$; otherwise it is the pair consisting of $N_\eta(v_K)$ and the value of $v_{\mathcal{P}}$ in $\mathcal{P}$. This is at present a partial test. For the second type of code, fix a finite monoid generating alphabet Z for $\hat{G}$, including the inverses of a finite group generating set, and test words by their values in $\hat{G}$. This second test is total by the hypothesis of Theorem 6.2.

Choose computably an integer $D_0 > M$ such that $2^{D_0-1} \geq \max\{|Y|, |Z|\}$. Encode the letters of Y , in their chosen order, by distinct words beginning with $\mathbf{b}_T$ and followed by $D_0 - 1$ letters from $\{\mathbf{a}_T, \mathbf{c}_T\}$, in lexicographic order. Encode Z in the same way using the other triple. Write $C(v)$ for concatenation of the appropriate codes and put $C(\varepsilon) = \varepsilon$. The type of a nonempty code sequence is unambiguous. Codes preserve shortlex comparison and have length D_0 per generator.

Enumerate all nonempty finite ordered tuples

$$(w_1^{(j)}, \dots, w_{s_j}^{(j)}) \quad (j \geq 1)$$

of distinct Σ -words avoiding ζ ; the empty word is allowed as a member. Choose integers N_j recursively so that

$$N_j > j, \quad N_j > \max_i |w_i^{(j)}|, \quad N_j > 2N_h + 2 + \max_i |w_i^{(h)}| \quad (h < j). \quad (34)$$

These inequalities concern *all potential* earlier envelopes, regardless of which rules will be selected. The increasing sequence N_j is computable without performing any tested-class evaluation.

We now define a partial procedure $F(\eta, w)$. Put $n = |w|$ and form the following finite rule table $\mathcal{R}_n$.

1. Include $\zeta a \rightarrow \zeta$ and $a\zeta \rightarrow \zeta$ for every letter $a \in \Sigma$. For each nonempty Y -word v with $D_0|v| \leq n$, test its class. If it is zero, include $C(v) \rightarrow \zeta$. Otherwise find the shortlex least word v' of the same tested class among all words of length at most $|v|$, including the empty word; include $C(v) \rightarrow C(v')$ if $v' \neq v$. Do the same for Z -words using equality in $\hat{G}$, with no zero case. Call these the *static rules*.
2. Process the tuples with $N_j \leq n$ in order. If every member of tuple j has no reducible occurrence under the static rules and previously selected rules, include

$$\mathbf{d}1^{N_j} w_i^{(j)} \mathbf{r}^{N_j} \mathbf{h} \longrightarrow \begin{cases} \varepsilon, & i = 1, \\ \zeta, & i > 1. \end{cases} \quad (35)$$

Some of these left sides may be longer than n ; they simply cannot act on the input.

3. Reduce w until no rule applies, using a fixed deterministic choice of occurrence and rule.

Whenever the tests terminate, the last step terminates: every rule decreases shortlex. In particular a nonempty code reduced to ζ shortens, since $D_0 > 2$. All other non-code rules shorten as well. Thus F is uniformly partial computable.

7.3 The fixed point and its length induction

Effective substitution of parameters gives a total computable index transformation $\tau(\eta)$ for the function $F(\eta, -)$. Kleene's Recursion Theorem [12] supplies an index with

$$f_\eta(w) = F(\eta, w) \quad (36)$$

as partial functions. Here is the short parameterization argument. Let U be universal evaluation and let $d(a)$ be an effectively obtained index for the partial function $w \mapsto U(U(a, a), w)$. Choose an index b computing the total function $a \mapsto \tau(d(a))$. For $\eta = d(b)$,

$$U(\eta, w) = U(U(b, b), w) = U(\tau(d(b)), w) = F(\eta, w).$$

Only now fix the presentation alphabet and code length belonging to this particular η . There is no advance bound on the machine's size or working space. Write $f = f_\eta$ and $K = K_\eta$.

Lemma 7.2. *The function f is total and length-nonincreasing. The rule tables are stable on bounded-length words: for $k \leq n$, $\mathcal{R}_k$ and $\mathcal{R}_n$ have the same rules whose left sides have length at most k . Each $\mathcal{R}_n$ is confluent on words of length at most n , and*

$$f(xf(w)y) = f(xwy) \quad (x, w, y \in \Sigma^*). \quad (37)$$

Proof. We prove all bounded assertions together by induction on n . For $n = 0$, there are no tested classes or tuple decisions, and the empty word is unchanged. Suppose $n > 0$ and the assertions are known at smaller bounds. Every use of N_η in a static test, including each candidate in a least-representative search, uses a Y -word of length at most $\lfloor n/D_0 \rfloor$. Its K -projection has grade at most

$$m = M \lfloor n/D_0 \rfloor < n. \quad (38)$$

The induction hypothesis supplies (30)–(31) at m . Lemma 7.1 therefore makes all these tests terminate. There are only finitely many tests and tuple decisions, and the subsequent shortlex-decreasing reduction terminates.

We first establish stability. For a fixed generator word v , its static rule is determined by the same tests and the same candidates of length at most $|v|$ at every larger bound. Consider a tuple with $N_j \leq k \leq n$. Each tested body has length less than N_j , hence less than k . New static left sides longer than k cannot occur in it. Induction over earlier tuple numbers shows that the previously selected rules that can occur in the bodies also agree. Hence the decision for tuple j agrees at both bounds. A new tuple with $N_j > k$ has every selector side longer than k , so cannot add a rule on a word of length at most k . This proves stability, including the decisions for selected rules too long to act at the smaller bound.

It remains to prove local confluence for $\mathcal{R}_n$. All reductions stay inside the length-at-most- n ball. Disjoint occurrences can be reduced in either order. If the word contains ζ , every branch still contains it: absorption retains ζ , and no other rule has a left side containing that letter. Absorption joins all branches at ζ . We check the other overlaps explicitly.

Selectors from different stages. Let the older left side have total length L , and let the newer one be $\text{dl}^N \text{wr}^N \text{h}$. By (34), $L < N$ and $|w| < N$. A nonempty old suffix cannot equal a new prefix: the

former ends in $\mathbf{h}$, while a prefix of length at most L lies in the initial $\mathbf{d}1^N$. An old prefix cannot equal a new suffix, since their first letters are respectively $\mathbf{d}$ and either $\mathbf{r}$ or $\mathbf{h}$. These exclude both orientations of a proper boundary overlap. For containment, an old occurrence cannot start at the new initial $\mathbf{d}$, because its terminal $\mathbf{h}$ would lie in the initial $\mathbf{1}$ -run. Any other start must be in the body. It cannot leave the body: it cannot end in the final $\mathbf{r}$ -run, and reaching the final $\mathbf{h}$ would require crossing a run longer than the entire old side. Thus it would lie wholly in the body, contrary to the newer tuple's irreducibility test.

Selectors from one stage. A second start strictly inside one side must lie in its body. There are then fewer than N body letters left before the final $\mathbf{r}$ -run, so that start cannot be followed by N consecutive $\mathbf{1}$'s. If two sides start at the same place and their body lengths differ, the terminal $\mathbf{h}$ of the shorter falls in the final $\mathbf{r}$ -run of the longer, because both body lengths are less than N . Equal body lengths force equal bodies, hence the same rule, as tuple members are distinct. This also covers an empty body.

A code and a selector. Code letters are disjoint from the envelope markers. An overlapping code side would therefore have to lie wholly inside the selector's body. The static irreducibility test excluded this. All static sides short enough to occur there were already present when the tuple was processed, because the body's length is less than N_j .

Two code sides. They have the same type and aligned starts: within a code sequence, $\mathbf{b}_T$ or $\mathbf{b}_G$ occurs only at a block boundary. Their union is therefore an entire code sequence $C(v)$, of length at most n . For group codes, a replacement preserves the value of this union by group multiplication. For product codes, replacing a subword by a nonzero tested-equal word preserves the tested class of the union by (32) and equality in $\mathcal{P}$. Indeed each old or new union has generator length at most $\lfloor n/D_0 \rfloor$, and hence its K -projection has grade at most m . We use this bound separately for both endpoints: a shorter generator word need not have smaller actual K -grade.

If a subword tests zero, apply (32) to its K -projection and Ω , whose grade is zero. The contextual word containing Ω has output Ω , so the union also tests zero. Hence if one branch introduces ζ , it reduces the whole union to ζ , and the other either does the same or retains a zero-class code sequence, to which its whole-word zero rule applies. That remaining sequence cannot be empty, since $N_\eta(\varepsilon) = \varepsilon \neq \Omega$. The same argument covers a zero-class union when both first replacements are code sequences.

If the union class is nonzero, let v_* be its least representative among words of length at most $|v|$. Either branch yields a word v_b of the same class with $|v_b| \leq |v|$. Since v_b is one of the candidates defining v_* , we have $|v_*| \leq |v_b|$, and v_* is also the least candidate within this smaller bound. Each branch therefore reduces by its whole-word rule to $C(v_*)$, unless it already equals that word. The representative v_* may be empty; no empty left side is needed. These joining reductions can be made on the union before acting on any newly adjacent outside letters.

This exhausts the local peaks. Termination and local confluence give confluence on the ball by Newman's Lemma [16]. Indeed, by well-founded induction, assume every proper descendant of a word has a unique normal form. Any two first steps have a common descendant by local confluence; the induction hypothesis makes their normal forms equal, proving uniqueness for the original word and hence confluence. Finally, if $|xwy| \leq n$, perform inside $x(-)y$ the normalization of w computed at bound $|w|$. By stability every rule used belongs to $\mathcal{R}_n$, and length never increases. Confluence gives $f(xf(w)y) = f(xwy)$. This establishes the budget hypotheses at n and completes the induction. $\square$

7.4 The two monoids and their embeddings

Let H be the monoid on Σ whose equations are the union of all the rule tables. Every word is related to its computed normal form, which is irreducible for this union by stability. Conversely, a contextual use of any one equation preserves that form: choose a bound containing both endpoints and the equation, and apply stability and confluence from Lemma 7.2. Thus two words represent the same element of H if and only if their literal f -outputs agree. This proves decidability. The empty word and the single letter ζ are distinct irreducibles; they represent the identity and the absorbing zero of H .

The budget hypotheses now hold at every length, so Lemma 7.1 gives an equality test for K . In particular its identity and zero are distinct. The tested product classes are now the actual elements of

$$T = ((K \setminus \{\Omega\}) \times \mathcal{P}) \cup \{0_T\},$$

with coordinatewise multiplication and every pair having first coordinate Ω collapsed to zero. Here a letter of X_η acts in the first coordinate and a generator of $\mathcal{P}$ in the second; the letter Ω maps to 0_T . Thus Y generates T .

For a nonzero element of T , take its least shortlex Y -representative v . The code $C(v)$ is irreducible: a subcode rule would either force the element to be zero or produce a smaller representative; no selector or absorption side occurs in a pure code word. Every other representative reduces by its whole-word rule to this same code, and every zero-class code reduces to ζ . Consequently coding defines an injective zero-preserving monoid map

$$\kappa : T \longrightarrow H.$$

It is unital because the least representative of the identity is empty, and it is multiplicative because codes concatenate. The identical argument for Z -words embeds $\widehat{G}$ unitaly in H . Its image consists of units, since the products of mutually inverse group words have the empty representative. This includes the trivial group and does not require a nonempty generating set.

Let $z \in H$ be represented by the separator $\mathbf{z}$. For least representatives v, v' of nonzero elements of T , the words

$$C(v)\mathbf{z}C(v') \tag{39}$$

are irreducible. No code crosses the separator; each code portion is already irreducible; and there are no selector markers or ζ . These literal words recover the ordered pair, including when one or both representatives are empty. They are nonzero because none equals ζ . This proves the separator assertion of Theorem 6.2.

7.5 Selectors and multiplication by raw words

Take any ordered tuple of distinct nonzero elements $h_1, \dots, h_s$ of H , and let $w_1, \dots, w_s$ be their normal words. These words avoid ζ , since any word containing it reduces to the zero word. Their tuple occurs in the enumeration, say at stage j . A word irreducible for the final rule set is irreducible for the static and selected rules present at stage j . Hence this tuple passes its test. With

$$L = \mathbf{dl}^{N_j}, \quad R = \mathbf{r}^{N_j}\mathbf{h}$$

as elements of H , the selected equations give $Lh_1R = 1$ and $Lh_iR = \zeta$ for $i > 1$. The argument includes a tuple containing the identity, whose normal representative is empty.

Finally define a multiplicative map in the opposite direction by

$$c : H \longrightarrow K, \quad c([w]) = [\text{raw}(f(w))].$$

The normal-form characterization of H makes this well-defined. Every raw singleton is its own literal N_η -output. Distinct normal words therefore give distinct elements of K , and none of these elements is its identity or its zero. For normal words u, v , (33) gives

$$c([u])c([v]) = [\text{raw}(f(uv))] = c([uv]).$$

Thus c is injective and multiplicative, as required. It is not unital: even $c(1) = [\text{raw}(\varepsilon)]$ is distinct from 1_K . Also $c(\zeta) \neq \Omega$; it is absorbing only within $c(H)$. The finite presentation of K came from the compiler, while the finite alphabet and normal-form procedure give the claimed finite generation and decidability of H . All conclusions of Theorem 6.2 are proved.

A The finite compiler

We prove Lemma 7.1. The input is a program index η for a partial function $f_\eta : \Sigma^* \rightharpoonup \Sigma^*$ on the fixed finite alphabet Σ . We construct a finite monoid presentation and its partial evaluator. The main issue is to control arbitrary words in that presentation, including malformed machine configurations.

Encoding a finite machine by contextual equations, and checking the consequences of using those equations in reverse, are classical ideas [17, pp. 3–6]. Murskii used finite machine presentations to embed finitely generated recursively presented semigroups in finitely presented semigroups [15, Theorem 1.2 and Section 1.5]; his construction also controls malformed initial configurations [15, Section 2.4]. For the present construction we also need a single evaluator on all presentation words, normalization paths that never increase grade, and invariance under equations in arbitrary contexts within the grade bound. We prove these properties explicitly below.

Fix a deterministic one-tape machine for f_η , with tape cells indexed by the nonnegative integers. A left move at cell zero stays there. The input begins at cell zero and is followed by blanks; a returned string also begins at zero and ends at its first blank. We use an effective compilation of the acceptable programming system into this convention. The resulting finite transition table is part of the compiler’s input; constructing the presentation never runs the program.

A.1 Cells, paths, and the validator

A *cell* consists of an immutable base in $\Sigma \sqcup \{\mathbf{C}, \mathbf{D}, \mathbf{W}\}$ and a finite tag. Bases in Σ are called anchors. The other bases mark two cuts and a workspace cell. A tag has an anchor-read bit, a workspace-filled bit, a work symbol from the compiled machine’s finite tape alphabet, an output-reading bit, and an output mark in $\Sigma \sqcup \{\text{none}\}$. The *clean* tag has every bit unset, work symbol blank, and output mark none. All these fields may be included on every base; fields not used there are simply left unchanged. A bare cell base in a rule means the cell with its entire tag clean.

The presentation has a zero letter Ω and the arrows of the quiver in Figure 2. There are arrows $\alpha : O \rightarrow L$ and $\omega : R \rightarrow O$, head arrows $L \rightarrow R$, and two loop copies b_L, b_R of every tagged cell b , at L, R , respectively. There are no vertex-identity generators: the empty word is the global identity. Add the equations making Ω absorbing and making every incompatible pair of arrows equal to Ω .

A nonempty path is cut at every $\omega\alpha$ into *sectors*. Each contains at most one head, since returning from R to L requires such a cut. A headed sector has optional α , left loops, a head,

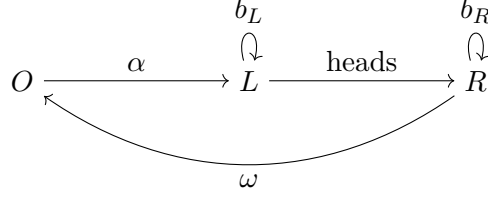


Figure 2: Nonzero words are paths. Cutting at each interior visit $\omega\alpha$ to O leaves sectors with at most one head.

right loops, and optional ω . Sectors without heads are allowed. Thus the same notation describes complete computations and words with missing boundaries.

Assign grade zero to Ω and workspace cells, grade two to cut cells with base $\mathbb{C}$, and grade one to all other generators. Tags do not affect grade. Extend grade additively to words, writing γ . The maximum generator grade is therefore $M = 2$. Workspace can be added without increasing grade, although it increases literal word length.

Use raw and quota heads $\mathbf{Q}, \mathbf{P}$, validator heads $\mathbf{I}, \mathbf{J}, \mathbf{J}', \mathbf{N}, \mathbf{V}$, a disjoint set of machine heads, and a cleanup head $\mathbf{Z}$. For an anchor word x , the notation x_L or x_R means its sequence of clean cell copies. A sector is *full raw* if it has the form

$$\text{raw}(x) = \alpha \mathbf{Q} x_R \omega, \quad x \in \Sigma^*.$$

The word x is its payload. To combine two such sectors, first impose

$$\mathbf{Q} a_R = a_L \mathbf{Q} \quad (a \in \Sigma), \quad (40)$$

$$\mathbf{Q} \omega \alpha \mathbf{Q} = \mathbf{C}_L \mathbf{P} \mathbf{D}_R, \quad \mathbf{P} \mathbf{D}_R = \mathbf{W}_L \mathbf{P} \mathbf{D}_R, \quad \mathbf{P} \mathbf{D}_R = \mathbf{I} \mathbf{D}_R. \quad (41)$$

In the following validator equations, the arrows specify the forward direction used to analyze the machine; the defining relations are equations in both directions:

$$\begin{aligned} \mathbf{W}_L \mathbf{I} &\longrightarrow \mathbf{I} \mathbf{W}_R, & \mathbf{C}_L \mathbf{I} &\longrightarrow \mathbf{J} \mathbf{C}_R, & a_L \mathbf{J} &\longrightarrow \mathbf{J} a_R, \\ \alpha \mathbf{J} &\longrightarrow \alpha \mathbf{J}', & \mathbf{J}' a_R &\longrightarrow a_L \mathbf{J}', & \mathbf{J}' \mathbf{C}_R &\longrightarrow \mathbf{C}_L \mathbf{N}, \\ \mathbf{N} \mathbf{W}_R &\longrightarrow \mathbf{W}_L \mathbf{N}, & \mathbf{N} \mathbf{D}_R &\longrightarrow \mathbf{D}_L \mathbf{V}, & \mathbf{V} a_R &\longrightarrow a_L \mathbf{V}, & \mathbf{V} \omega &\longrightarrow q_{\text{start}} \omega. \end{aligned} \quad (42)$$

The anchor rules range over $a \in \Sigma$.

An *entry* has head $\mathbf{I}$ immediately before clean $\mathbf{D}_R$. The validator reaches q_{start} from an entry only when that entry has the complete form

$$\alpha x_L \mathbf{C}_L \mathbf{W}_L^s \mathbf{I} \mathbf{D}_R y_R \omega, \quad x, y \in \Sigma^*, \quad s \geq 0. \quad (43)$$

Indeed the first left sweep checks precisely the clean workspace up to $\mathbf{C}$, and the next checks only clean anchors up to α . The rightward sweeps then check these anchors, the first cut, the workspace, the second cut, and only clean anchors up to ω . Thus both boundaries and every tag are checked. At the handoff all cells are on the left of the head.

These relations prepare a raw product for computation. Slide the first head across x , merge the two adjacent heads, insert s workspace cells, and enter the validator. The simulation and cleanup defined below will complete the following passage whenever $f_\eta(xy)$ is defined and has length at most $|xy|$, provided s is sufficiently large:

$$\begin{aligned} \text{raw}(x) \text{raw}(y) &\rightsquigarrow \alpha x_L \mathbf{C}_L \mathbf{W}_L^s \mathbf{I} \mathbf{D}_R y_R \omega \\ &\rightsquigarrow \text{raw}(f_\eta(xy)). \end{aligned} \quad (44)$$

The first passage is the preparation just described; the second is validation, simulation on the workspace, and cleanup. The two anchor blocks retain the input xy and provide space for its output letters.

Successful computations alone do not control the presented monoid, because each equation can also be used in reverse. The validator and simulator will preserve the ordered cell bases and have finitely many tags and head states. Consequently their transitions on any fixed sector form a finite graph, even if the simulated program never halts. We will search its whole weak component—allowing transitions in either direction—to normalize arbitrary configurations. The component analysis will distinguish those that contain an entry from those that do not. Quota insertion is kept outside this finite graph.

A.2 A finite transition table for the computation

We specify the remaining machine equations explicitly through finite scan instructions. This makes the construction uniform in η while avoiding a separate list of rules for each tape symbol.

Every state has one designated reading side. A right-reading state has, for each tested tagged cell, at most one rule of either form

$$qb_R \longrightarrow b'_L q' \quad \text{or} \quad qb_R \longrightarrow q' b'_R; \quad (45)$$

a left-reading state uses the mirrored forms. These instructions cross the cell or stay in the same gap, respectively. The cells b, b' always have the same base. Boundary tests are stationary rules $\alpha q \rightarrow \alpha q'$ or $q\omega \rightarrow q'\omega$ on the designated side. A failed test has no rule. A finite cell predicate expands into the list of tagged cells satisfying it; complementary cases are disjoint. Finite registers are incorporated into the state set by a finite Cartesian product.

Here are the scan instructions used below, together with their complete expansion convention.

- To rewind, cross every cell on the left unchanged. At α , change state without moving and start the specified right-reading continuation. If α is absent, the scan gets stuck at the left end of the word.
- To find the first cell satisfying a predicate, test cells on the right. At a failure cross unchanged and repeat. At a match use the specified stationary update and continuation. At ω , use the specified end continuation, or get stuck if failure was prescribed.
- To go to the right boundary, cross all right cells unchanged and use the specified stationary continuation at ω .

All invocations are expanded with distinct phase states, apart from explicit repetitions of a loop. In particular q_{start} is fresh: its only incoming rule is the validator handoff, and its only outgoing rule is $q_{\text{start}}\omega \rightarrow q_{\text{rewind}}\omega$, where q_{rewind} starts the first rewind. No continuation returns to q_{start} or to a validator state.

The following finite control program supplies all further internal machine equations.

1. *Copy the input.* Rewind, and find the first anchor whose read bit is unset. Set that bit, carry its base a in finite state, rewind, and find the first workspace cell whose filled bit is unset. Write a on its work track and set its filled bit. Failure to find that workspace cell is stuck. Rewind and repeat the anchor search.

When the anchor search reaches ω , rewind and check that an unfilled workspace cell remains; otherwise get stuck. Rewind once more and locate the first workspace cell for simulation.

On (43) this writes exactly xy in anchor order into the initial workspace cells, followed by a blank.

2. *Run the given machine.* The simulated head reads a workspace cell on its right. For each machine state and work symbol, expand its single transition using (45). A stationary move writes and stays. A right move writes and crosses the current cell, and the next simulation state requires another workspace cell on the right. A left move first writes without crossing, then enters a left-reading intermediate state. That state crosses an immediately left workspace cell unchanged, or stays if the immediately left base is C; any other case is stuck. It then returns to a right-reading simulation state. This implements the left-end convention. A halt is recognized only at a workspace cell in bounds.
3. *Read the output.* After halt, rewind and find the first workspace cell whose output-reading bit is unset. A blank work symbol ends the output. For a work symbol $a \in \Sigma$, set that bit, carry a in finite state, rewind, and place output mark a on the first anchor whose output mark is none. Failure to find such an anchor is stuck. Rewind and repeat. An invalid output symbol, or exhaustion of the workspace without a blank, is stuck. When output ends, go to ω and enter a fresh state q_{ok} , with no outgoing internal transition.

Each instruction uses only the finite predicates, finite registers, and updates just specified. Expanding all cases gives a finite list of equations computable from the machine table. Each state tests only its designated side and has disjoint cases. Consequently the internal forward transition relation has outdegree at most one on *every* sector, including sectors with missing boundaries and arbitrary tags. Every transition preserves the number and ordered bases of the cells.

On a passing entry, simulation success computes $f_\eta(xy)$ genuinely. Conversely, if $f_\eta(xy)$ is defined and has length at most $|xy|$, then some finite quota s permits success: take more workspace cells than the input length, the largest tape index visited by the halting computation, and the output length. The extra cell supplies its blank delimiter. There are $|xy|$ anchors for the output marks. This includes empty input: at least one workspace cell is supplied, and a length bound forces the output to be empty.

Add the exit equation $q_{ok}\omega \rightarrow Z\omega$ and the following cleanup equations. An anchor cell b with output mark $a \in \Sigma$ has

$$b_L Z \longrightarrow Z a_R. \quad (46)$$

For every other tagged cell use $b_L Z \rightarrow Z$, and add $\alpha Z \rightarrow \alpha Q$. Cleanup is deterministic and never increases grade, even on dirty data. Applied to a successful passing entry, it leaves

$$\alpha Q f_\eta(xy)_R \omega. \quad (47)$$

The leftward sweep places successive output letters on its right in the correct order. All unused anchors and all nonanchor cells disappear.

A.3 Finite components on arbitrary sectors

We now analyze sectors with arbitrary tags and optional boundaries. For a validator or machine sector, form the graph of internal forward transitions, excluding entry from P and exit to Z. We use its entire weak connected component. The ordered bases of its cells and its boundary flags are fixed. If there are k cells and h internal head states, with at most c tags per cell, there are at most $(k+1)hc^k$ possible configurations with these data. Its weak component can therefore be found by a finite search through elementary compiled steps. This search does not call f_η or wait

for a computation to halt. In particular, the quota insertion rule is excluded: finite grade alone would not bound the number of grade-zero workspace cells.

Lemma A.1. *Every internal weak component has at most one entry and at most one sector whose head is q_{ok} immediately before its final ω . If both occur, the forward computation from the entry reaches that sector and has the clean form and genuine output described in (43) and (47).*

Proof. An entry has internal indegree zero: the only possible predecessor of an I-sector is a left sweep, whose output has clean W immediately right of the head, whereas an entry has clean D. Every other validator sector has indegree at most one. For head J, its possible predecessor types are distinguished by clean C versus an anchor immediately right. For J' they are distinguished by α versus an anchor immediately left; for N by clean C versus W immediately left; and for V by clean D versus an anchor immediately left. There are no transitions from machine states to validator states.

The validator phases progress in their displayed order, and each sweep is finite. An entry whose validation fails before handoff therefore belongs to an isolated finite chain: its vertices have at most one predecessor and one successor, the entry has no predecessor, and the failed end has no successor. It shares its component with no other entry. Every passing entry has (43). Its fixed base sequence and boundary flags determine it uniquely, including its clean tags and the head gap immediately before the unique D. Hence there is at most one entry in any component.

Finally, a finite directed graph with outdegree at most one has a single eventual cycle or terminal vertex in each weak component. Indeed the unique forward trajectories of adjacent vertices have the same eventual object, and this propagates along weak paths. A sector ending in $q_{ok}\omega$ is terminal. Thus it is the only such sector in its component, and every forward trajectory there reaches it. If an entry is present, it therefore passes validation and genuinely succeeds. $\square$

A.4 The partial evaluator

The component lemma leaves a choice when both an entry and a successful terminal occur. Returning to the entry, removing its quota, and undoing the merge, then sliding the raw heads left, gives $\text{raw}(x) \text{raw}(y)$; cleaning up the terminal gives $\text{raw}(f_\eta(xy))$. We give the entry priority, so that internal equations have identical preprocessing outputs. The final combination of consecutive raw sectors will identify the two routes in (44), using the contextual identity assumed in Lemma 7.1 when other raw sectors are adjacent.

This priority also applies to a failed or malformed entry; in that case quota removal need not produce two full raw sectors. Conversely, a successful component without an entry need not encode a genuine input computation. It can use its unique terminal cleanup directly. The following algorithm includes all these cases.

The algorithm N_η preserves the empty word. It sends every word containing Ω , and every nonpath, to Ω , using the zero relations. For a nonempty path, first preprocess its sectors as follows.

1. Leave sectors without heads unchanged. Move Q to its leftmost reachable gap using (40), across clean anchors only.
2. If P has clean D_R immediately right, remove all immediately preceding clean W_L by the quota equation. If clean C_L is then immediately left, unmerge $C_L P D_R$ into $Q \omega \alpha Q$ and normalize the two resulting raw heads as above. Otherwise keep the trimmed sector. A P without immediately right clean D_R is kept.
3. Run Z cleanup until it is stuck or becomes Q; in the latter case normalize that raw head.

4. For any validator or machine head, find its finite internal weak component. If there is an entry, move to it along a weak path, revert to P , and use the preceding rule. If there is no entry but there is a sector ending in $q_{ok}\omega$, move there and use cleanup. If neither exists, use the least sector in the component in shortlex order.

Lemma A.1 makes the choices in the last rule unambiguous. The ordering is fixed independently of any grade budget. All these operations preserve the original outer boundaries; only unmerge creates an interior $\omega\alpha$ and splits a sector.

Replace each maximal run of $t \geq 2$ consecutive full raw sectors, with payloads $u_1, \dots, u_t$, by

$$\alpha Q f_\eta(u_1 \cdots u_t)_R \omega. \quad (48)$$

A run of length one is left unchanged, without evaluating f_η . This completes the definition of N_η . Its only calls to that function are in (48).

Assume now the budget hypothesis of Lemma 7.1 at m . Every preprocessing operation has a derivation that never raises grade. Internal component paths preserve every base and hence grade. Raw slides and entry reversal preserve grade. Quota cells have grade zero. Unmerge preserves grade because

$$\gamma(C_L P D_R) = 2 + 1 + 1 = 1 + 1 + 1 + 1 = \gamma(Q \omega \alpha Q).$$

Cleanup preserves or decreases it. Each argument in (48) has length at most the original total grade, since every anchor contributes one. Thus all required calls terminate for an input of grade at most m .

The replacement of two full raw sectors is also derivable without raising grade. Move the first head right across its anchors, merge at $Q \omega \alpha Q$, insert a sufficient grade-zero workspace quota, enter, simulate successfully, and clean up. The result is the raw word for $f_\eta(u_1 u_2)$. Repeating this operation gives (48), because the budget hypothesis implies

$$f_\eta(f_\eta(u_1 u_2) u_3) = f_\eta(u_1 u_2 u_3)$$

and the analogous identities for longer runs. The successive arguments to f_η have length at most $|u_1| + \cdots + |u_t|$, because f_η is length-nonincreasing. Each elementary derivation step preserves or decreases grade, including the insertion of grade-zero workspace. Thus every call and every intermediate word remains within its required budget. This proves termination and the asserted grade-nonincreasing normalization path.

A.5 Invariance under equations and contexts

We verify that one contextual equation step preserves N_η when both endpoints have grade at most m . All nonzero equation sides are nonempty paths with the same initial and terminal vertices. This includes the deletion $b_L Z \rightarrow Z$. A replacement therefore preserves whether the whole word is a path. Zero equations and nonpaths give Ω on both sides, so it remains to compare paths, using their sectors in the whole contextualized word.

Raw slides have the same leftmost reachable gap. Quota changes have the same trimmed sector. Entry from P has the same preprocessing on both sides, since the internal component gives priority to its entry, even if that entry is malformed and validation fails. Internal validator or machine equations leave the weak component unchanged. Cleanup equations give the same result by forward determinism, including the last step to Q . A merge is immediately undone by quota preprocessing before normalizing its two raw heads. This remains true when the two sectors have

incomplete outer boundaries or dirty cells. In each case the preprocessed sector list is identical, so its subsequent raw-run replacements agree.

The only remaining equation is $q_{\text{ok}}\omega \rightarrow Z\omega$. If its internal component has no entry, both sides use the same unique terminal cleanup. If it has an entry, Lemma A.1 says that the entry is (43) and its computation returns $f_\eta(xy)$. Entry-priority preprocessing on one side gives two full raw sectors for x, y ; cleanup on the other gives a single full raw sector for $f_\eta(xy)$. With no full raw neighbors, the pair is replaced by exactly that singleton. Otherwise let a, b be the concatenations of the neighboring raw words on the left and right. The final runs have outputs

$$f_\eta(axyb) \quad \text{and} \quad f_\eta(af_\eta(xy)b),$$

which agree by the budget hypothesis. The anchor count $|axyb|$ on the entry side is at most the original grade, so this use is legitimate. Passing validation forces both α and ω ; an incomplete sector cannot occur in this exceptional comparison. This proves equation-step invariance in every context.

If $N_\eta(u) = N_\eta(v)$ and both xuy, xvy have grade at most m , transplant the nonincreasing normalization paths of u and v into $x(-)y$. Additivity keeps every intermediate word within budget. Equation-step invariance identifies each input's answer with the answer of its contextualized normal form. Those two middle strings are literally equal, proving bounded context compatibility. This argument does not require sector boundaries to survive the attachment of context.

If the hypotheses hold at every budget, every finite derivation is covered by some budget. Invariance then implies that equal monoid words have equal N_η answers. The converse follows from each word's derivation to its answer. Thus N_η is an exact word-problem evaluator. Its definition always leaves each raw singleton $\alpha Qu_R\omega$ literal, distinct from Ω and from the empty word; on two such sectors it gives the raw encoding of $f_\eta(uv)$ whenever the relevant budget hypothesis holds. All parts of Lemma 7.1 follow.

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